

Aerospace Academy Version

# Unit 7

## Final Engineering Design Challenge

### Unit Overview

Students complete a capstone-style aerospace design challenge using the full engineering process: problem definition, research, concept development, CAD, documentation, prototyping, testing, re-design, and final design review.

### Essential Question

How can engineers use the full design process to solve a realistic aerospace challenge?

Name: \_\_\_\_\_ Period: \_\_\_\_\_ Date: \_\_\_\_\_

**Learning Targets**

1. Explain the major concepts of final engineering design challenge.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

**Unit Deliverables**

- Project proposal
- Research notes
- Concept sketches
- Decision matrix
- CAD model or prototype plan
- Manufacturing documentation
- Prototype/test evidence
- Redesign proposal
- Final portfolio
- Final design review

**Vocabulary**

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| Term                    | Definition |
|-------------------------|------------|
| Capstone                |            |
| Design Brief            |            |
| Client Need             |            |
| Criteria                |            |
| Constraints             |            |
| Portfolio               |            |
| Prototype               |            |
| Test Plan               |            |
| Data Analysis           |            |
| Redesign                |            |
| Risk                    |            |
| Design Review           |            |
| Technical Documentation |            |
| Reflection              |            |

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## Unit Project: Final Aerospace Design Challenge

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

**Day 1 — Final Challenge Launch**

**Objective:** I can explain the final design challenge and expectations.

**Bellwork:** What skills from this course will you need most?

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**Core Task:** Review challenge options and choose or receive a mission prompt.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What mission are you considering?

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**Day 2 — Problem Definition**

**Objective:** I can define the final design problem clearly.

**Bellwork:** Why should the final problem be specific?

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**Core Task:** Write mission objective, stakeholders, criteria, constraints, and problem statement.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What is your most important criterion?

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**Day 3 — Research Plan**

**Objective:** I can plan research for a design challenge.

**Bellwork:** What information do you need before designing?

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**Core Task:** Identify research questions and begin collecting notes.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What did you learn today?

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**Day 4 — Research Synthesis**

**Objective:** I can summarize research into design requirements.

**Bellwork:** How does research become design criteria?

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**Core Task:** Create a research summary and design requirement list.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What requirement came from research?

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**Day 5 — Brainstorming**

**Objective:** I can generate multiple concepts for a final project.

**Bellwork:** Why should final projects begin with multiple ideas?

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**Core Task:** Generate eight ideas and select three for development.

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**Evidence of Work / Notes:**

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**Exit Ticket:** Which idea is most promising?

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**Day 6 — Concept Sketching**

**Objective:** I can create annotated sketches for three concepts.

**Bellwork:** What must your sketches communicate?

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**Core Task:** Create three annotated concept sketches.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What concept seems most feasible?

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**Day 7 — Decision Matrix**

**Objective:** I can choose a concept using evidence.

**Bellwork:** How will your team avoid choosing based on opinion only?

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**Core Task:** Score concepts using a decision matrix and select a final concept.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What concept won and why?

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**Day 8 — Design Proposal**

**Objective:** I can write a proposal for the selected design.

**Bellwork:** What should be approved before building?

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**Core Task:** Create a proposal with problem, concept, criteria, risks, and next steps.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What risk needs teacher approval?

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**Day 9 — CAD/Prototype Planning**

**Objective:** I can plan modeling and fabrication work.

**Bellwork:** What parts or files are needed?

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**Core Task:** Create part list, file plan, and fabrication plan.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What will you model first?

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**Day 10 — Build/Model Day 1**

**Objective:** I can begin building or modeling my final design.

**Bellwork:** What is today's measurable goal?

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**Core Task:** Work on CAD or prototype and document progress.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What did you complete?

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**Day 11 — Build/Model Day 2**

**Objective:** I can continue building or modeling with documentation.

**Bellwork:** What issue slowed progress yesterday?

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**Core Task:** Continue design work and record changes.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What changed?

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**Day 12 — Build/Model Day 3**

**Objective:** I can refine the design based on constraints.

**Bellwork:** What constraint is most challenging?

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**Core Task:** Continue design work and check against criteria.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What criterion is your design meeting?

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**Day 13 — Technical Documentation**

**Objective:** I can create technical documentation for the final design.

**Bellwork:** What documents prove your design is buildable?

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**Core Task:** Create drawings, BOM, notes, or assembly documentation.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What document is incomplete?

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**Day 14 — Test Plan**

**Objective:** I can create a test plan for my final design.

**Bellwork:** How will you know if the design works?

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**Core Task:** Write objective, variables, procedure, data table, and success criteria.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What will you measure?

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**Day 15 — Testing Day**

**Objective:** I can test a prototype or model and record data.

**Bellwork:** What makes testing fair?

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**Core Task:** Run tests or simulations and record results.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What did the data show?

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**Day 16 — Data Analysis**

**Objective:** I can analyze results and identify improvements.

**Bellwork:** What does the data suggest?

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**Core Task:** Create charts/tables or written analysis and identify failure points.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What needs redesign?

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**Day 17 — Redesign**

**Objective:** I can improve a design based on evidence.

**Bellwork:** Why is redesign expected?

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**Core Task:** Create and document a redesign proposal or revision.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What evidence supports your change?

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**Day 18 — Portfolio Work Day**

**Objective:** I can organize final project evidence.

**Bellwork:** What should a final portfolio prove?

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**Core Task:** Assemble problem definition, research, sketches, decisions, CAD, testing, and redesign.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What portfolio section is strongest?

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**Day 19 — Design Review Preparation**

**Objective:** I can prepare a professional final design review.

**Bellwork:** How should engineers communicate results?

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**Core Task:** Create final presentation slides and rehearse.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What slide needs work?

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**Day 20 — Final Design Review**

**Objective:** I can present and defend a final aerospace design solution.

**Bellwork:** What does professional presentation look like?

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**Core Task:** Present final design review and respond to questions.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What question was hardest?

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**Day 21 — Final Reflection**

**Objective:** I can reflect on my growth as an engineering student.

**Bellwork:** What skill improved most this year?

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**Core Task:** Complete final reflection and course debrief.

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**Evidence of Work / Notes:**

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**Exit Ticket:** What are you proud of?

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## Unit 7 Submission Checklist

| Item                        | Complete?                |
|-----------------------------|--------------------------|
| Project proposal            | <input type="checkbox"/> |
| Research notes              | <input type="checkbox"/> |
| Concept sketches            | <input type="checkbox"/> |
| Decision matrix             | <input type="checkbox"/> |
| CAD model or prototype plan | <input type="checkbox"/> |
| Manufacturing documentation | <input type="checkbox"/> |
| Prototype/test evidence     | <input type="checkbox"/> |
| Redesign proposal           | <input type="checkbox"/> |
| Final portfolio             | <input type="checkbox"/> |
| Final design review         | <input type="checkbox"/> |

## Unit 7 Project Rubric

| Category                                       | Points     |
|------------------------------------------------|------------|
| Technical accuracy and completeness            | 20         |
| Engineering process and documentation          | 20         |
| Aerospace connection and design intent         | 15         |
| Quality of sketches, models, data, or drawings | 20         |
| Use of evidence, testing, or decision making   | 15         |
| Reflection and professionalism                 | 10         |
| <b>Total</b>                                   | <b>100</b> |

## Unit 7 Student Work Templates

## Final Project Problem Definition

**Purpose:** Capstone planning

**Directions:** Define the final design challenge clearly before choosing a solution.

## Mission need

## Users / stakeholders

### Primary objective

### Key criteria

## Major constraints

### Success evidence

### Problem statement draft:

# Research Notes Template

**Purpose:** Research without copying

**Directions:** Record useful ideas from existing solutions. Focus on features, constraints, materials, failure points, and trade-offs.

**Topic/source:** \_\_\_\_\_

**Key idea:** \_\_\_\_\_

**Useful features:** \_\_\_\_\_

**Limits/risks:** \_\_\_\_\_

How this could inspire our design:

**Notes:**

## Research Notes Template

**Purpose:** Research without copying

**Directions:** Record useful ideas from existing solutions. Focus on features, constraints, materials, failure points, and trade-offs.

**Topic/source:** \_\_\_\_\_

**Key idea:** \_\_\_\_\_

**Useful features:** \_\_\_\_\_

**Limits/risks:** \_\_\_\_\_

How this could inspire our design:

**Notes:**

## Final Concept Sketch 1

**Purpose:** Concept development

**Directions:** Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

**Concept title:** \_\_\_\_\_

**Design goal:** \_\_\_\_\_

**Key features:** \_\_\_\_\_

### Initial Capstone Concept 1

**Strongest** \_\_\_\_\_

**feature:** **Biggest concern:** \_\_\_\_\_



## Final Concept Sketch 2

**Purpose:** Concept development

**Directions:** Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

**Concept title:** \_\_\_\_\_

**Design goal:** \_\_\_\_\_

**Key features:** \_\_\_\_\_

### Initial Capstone Concept 2

**Strongest** \_\_\_\_\_

**feature: Biggest concern:** \_\_\_\_\_



## Final Concept Sketch 3

**Purpose:** Concept development

**Directions:** Use quick, clear shapes. Add labels, arrows, and notes that explain function, materials, safety, payload protection, or design intent.

**Concept title:** \_\_\_\_\_

**Design goal:** \_\_\_\_\_

**Key features:** \_\_\_\_\_

### Initial Capstone Concept 3

**Strongest** \_\_\_\_\_

**feature: Biggest concern:** \_\_\_\_\_



## Decision Matrix Template

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**Purpose:** Evidence-based selection

**Directions:** Choose criteria that connect to the mission. Score each concept honestly using evidence from research, sketches, constraints, and trade-offs.

| Criteria | Weight | Concept 1 | Concept 2 | Concept 3 |
|----------|--------|-----------|-----------|-----------|
|----------|--------|-----------|-----------|-----------|

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**Total**

**Selected concept:** \_\_\_\_\_

**Justification:**

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## Final Design Isometric Sketch

**Purpose:** Isometric sketching

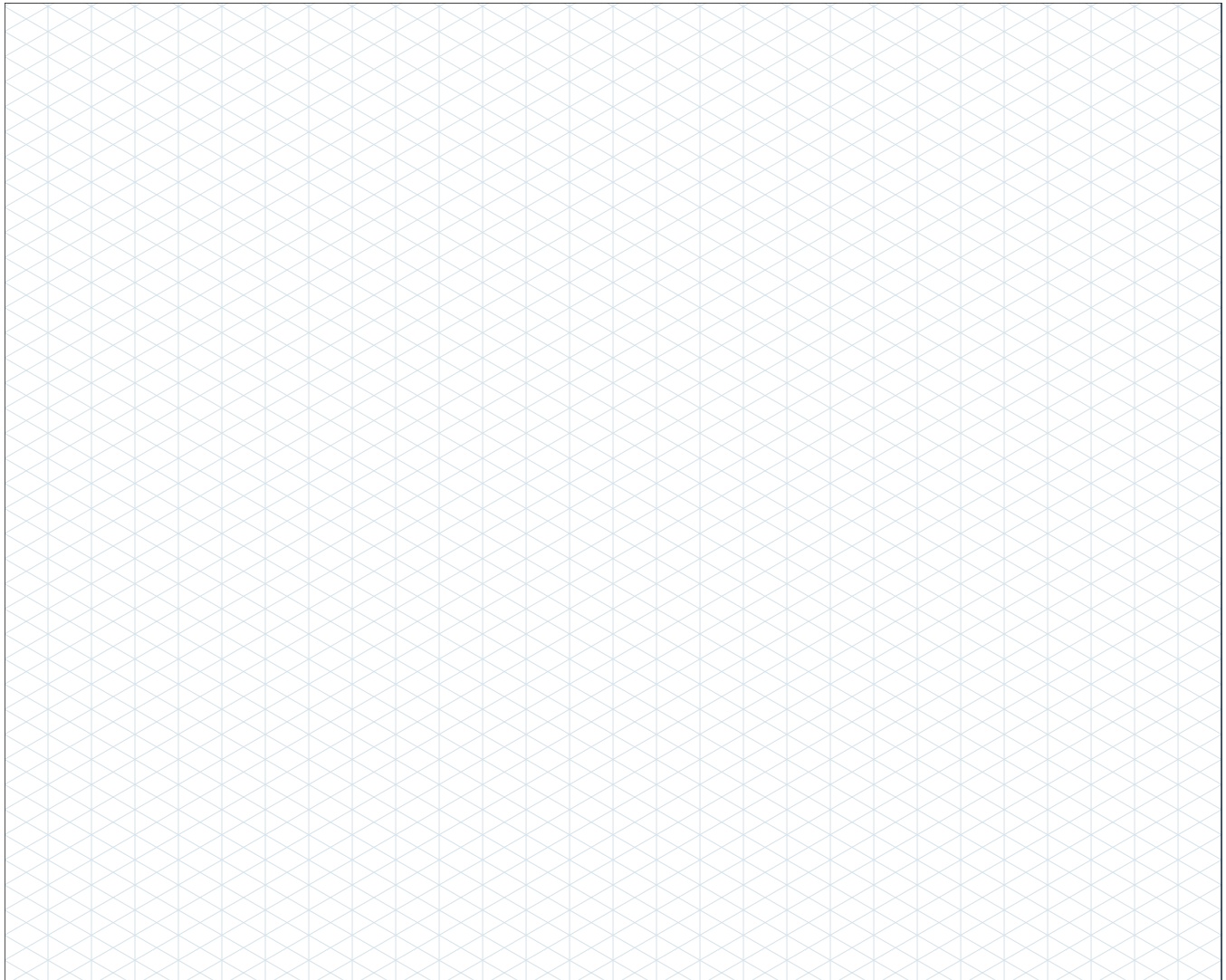
**Directions:** Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Purpose:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Notes / annotations:** \_\_\_\_\_

## Final Design Multiview Planning

**Purpose:** Orthographic / multiview sketching

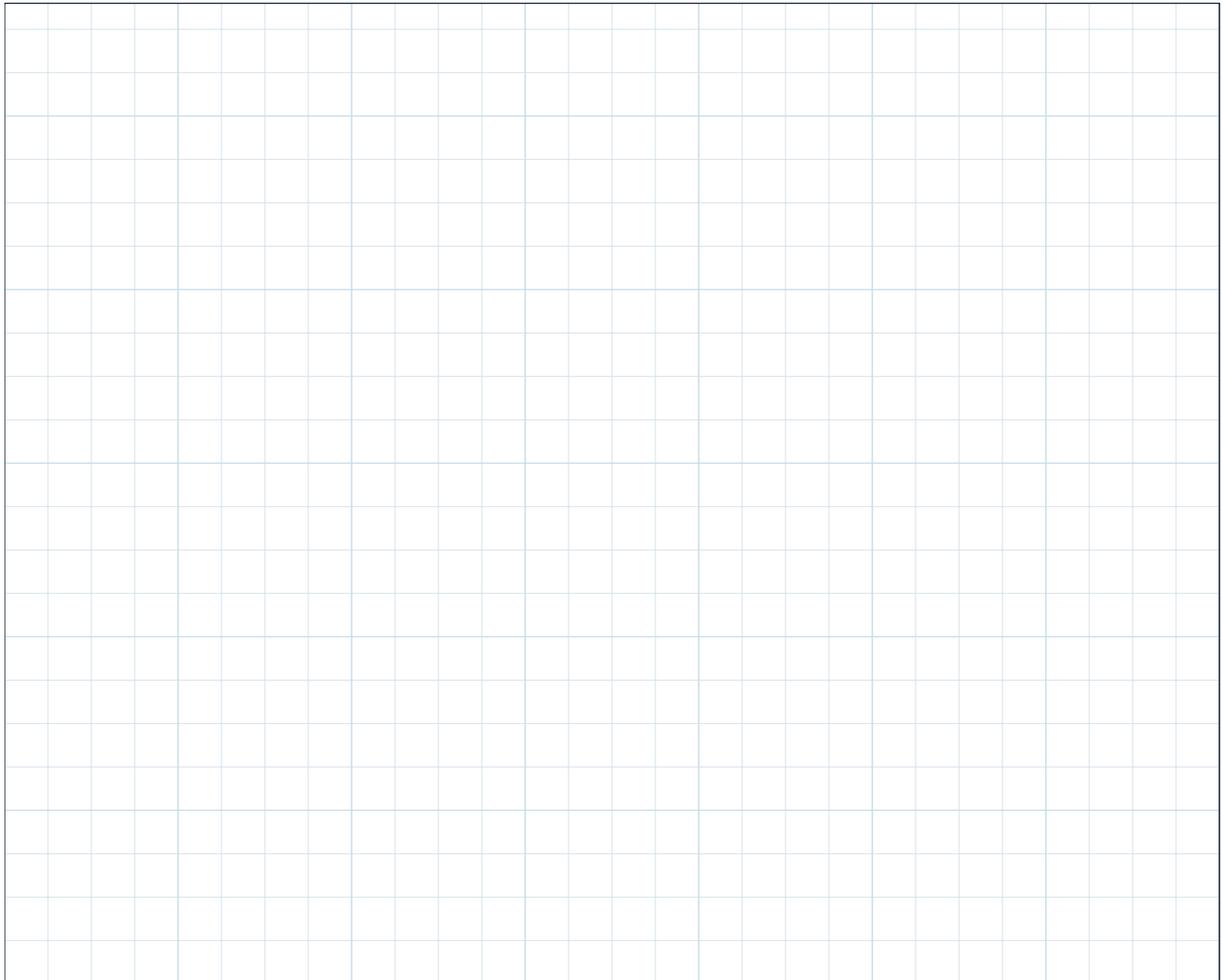
**Directions:** Use the orthographic grid to align the front, top, and right-side views. Show only the views requested. Use projection alignment and line conventions.

**Part name:** \_\_\_\_\_

**Scale:** \_\_\_\_\_

**Main view:** \_\_\_\_\_

**Date:** \_\_\_\_\_



**Dimension / line convention notes:** \_\_\_\_\_

## Final Prototype Testing Data

**Purpose:** Testing and analysis

**Directions:** Record measurements clearly. Include units, observations, failure points, and redesign decisions based on evidence.

| Trial | Measurement / Result | Observation | Change Needed |
|-------|----------------------|-------------|---------------|
| 1     |                      |             |               |
| 2     |                      |             |               |
| 3     |                      |             |               |
| 4     |                      |             |               |
| 5     |                      |             |               |

**Analysis / redesign notes:**

## Bill of Materials and Revision Template

**Purpose:** Documentation

**Directions:** List parts, materials, quantities, manufacturing methods, and revision notes so someone else could understand what changed.

| Item | Part Name | Material / Process | Qty. | Notes |
|------|-----------|--------------------|------|-------|
| 1    |           |                    |      |       |
| 2    |           |                    |      |       |
| 3    |           |                    |      |       |
| 4    |           |                    |      |       |
| 5    |           |                    |      |       |
| 6    |           |                    |      |       |

| Rev. | Date | Change Made | Reason for Change |
|------|------|-------------|-------------------|
| A    |      |             |                   |
| B    |      |             |                   |
| C    |      |             |                   |

## Final Design Review Planning

**Purpose:** Presentation planning

**Directions:** Plan a professional final design review. Include design decisions, evidence, limitations, and next steps.

| Slide | Required Message             | Evidence / Visual |
|-------|------------------------------|-------------------|
| 1     | Project title and mission    |                   |
| 2     | Problem statement            |                   |
| 3     | Criteria and constraints     |                   |
| 4     | Research and inspiration     |                   |
| 5     | Concepts and decision matrix |                   |
| 6     | CAD / drawings / prototype   |                   |
| 7     | Testing data                 |                   |
| 8     | Redesign and final solution  |                   |
| 9     | Reflection and next steps    |                   |

**Presenter notes:**

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